

Unit 11 - Justifying Relationships in Polygons With Circles

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The following B.E.S.T. Standards will be covered in this unit:

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships and theorems of triangles.

- *Clarification 1: Postulates, relationships and theorems include measures of interior angles of a triangle sum to 180° ; measures of a set of exterior angles of a triangle sum to 360° ; triangle inequality theorem; base angles of isosceles triangles are congruent; the segment joining midpoints of two sides of a triangle is parallel to the third side and half the length; the medians of a triangle meet at a point.*
- *Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flow chart proofs or informal proofs.*
- *Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.*

MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.

- *Clarification 1: Instruction includes using compass and straightedge, string, reflective devices, paper folding or dynamic geometric software.*

MA.912.GR.5.3 Construct the inscribed and circumscribed circles of a triangle.

- *Clarification 1: Instruction includes using a compass and straightedge, string, reflective devices, paper folding or dynamic geometric software.*

MA.912.GR.6.3 Solve mathematical problems involving triangles and quadrilaterals inscribed in a circle.

- *Clarification 1: Instruction includes triangles in a circle and semicircle.*

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.

- *Clarification 1: Within the Geometry course, instruction focuses on the connection to proofs within the course.*

MA.912.GR.5.4 (Honors only) Construct a regular polygon inscribed in a circle. Regular polygons are limited to triangles, quadrilaterals, and hexagons.

- *Clarification 1: When given a circle, the center must be provided.*
- *Clarification 2: Instruction includes using a compass and straightedge, string, reflective devices, paper folding or dynamic geometric software.*

Unit 11, Lesson 1: Constructing a Circumscribed Circle of a Triangle



Benchmark

MA.912.GR.5.3 Construct the inscribed and circumscribed circles of a triangle.



Learning Targets

- ☐ I can construct the circumscribed circle of a triangle.
- ☐ I can locate the circumcenter of a triangle.



11.1.1 Warm-Up: Lake Okeechobee and Deep Lake

Lake Okeechobee means "big water" in the language of the Seminole tribe. Lake Okeechobee has a surface area of 730 square miles. Deep Lake in Big Cypress National Preserve is a naturally occurring sinkhole that was formed roughly 6,000 years ago. Deep Lake has a surface area of 0.002 square miles.

1. How many Deep Lakes are needed to cover the same surface area as Lake Okeechobee?

Lake Okeechobee averages 9 feet in depth, whereas Deep Lake averages 95 feet in depth. Based on the volume of the two lakes it would take approximately 30,850 Deep Lakes to fill up Lake Okeechobee.

2. Explain why it takes more Deep Lakes to cover the surface area of Lake Okeechobee than to fill up Lake Okeechobee.



11.1.2 Exploration: Circumscribing A Triangle with A Circle

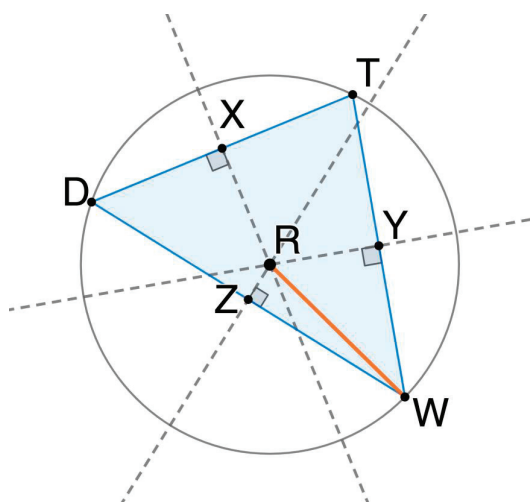


An **inscribed polygon in a circle** is a polygon which has all of its vertices on a circle.

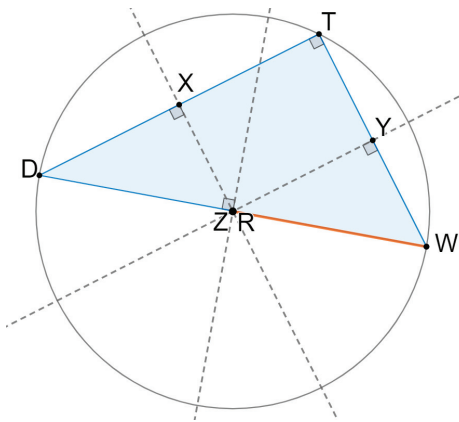
$\triangle TWD$ is inscribed in Circle R , where $\overline{RW}$ is a radius. Lines RX , RY , and RZ are perpendicular bisectors of $\overline{TD}$, $\overline{TW}$, and $\overline{DW}$, respectively.

Each diagram shows a different type of triangle inscribed in Circle R .

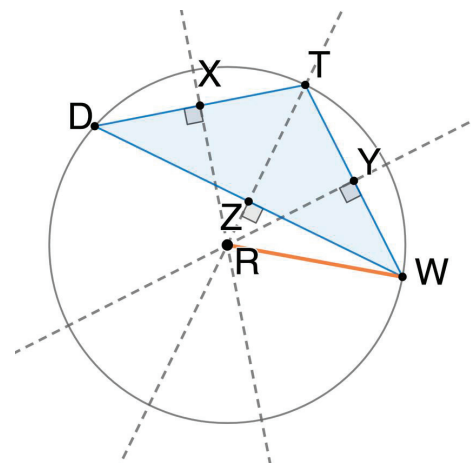
Acute



Right



Obtuse



- Determine similarities and differences among all three inscribed triangles.

Similarities	Differences

- Point R is the center of the circle that is circumscribed about $\triangle TWD$. Make a conjecture about point R as it relates to the perpendicular bisectors of the sides of $\triangle TWD$.

3. Ask two other pairs of partners for the conjectures they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

My Summary of Their Conjecture	Initials

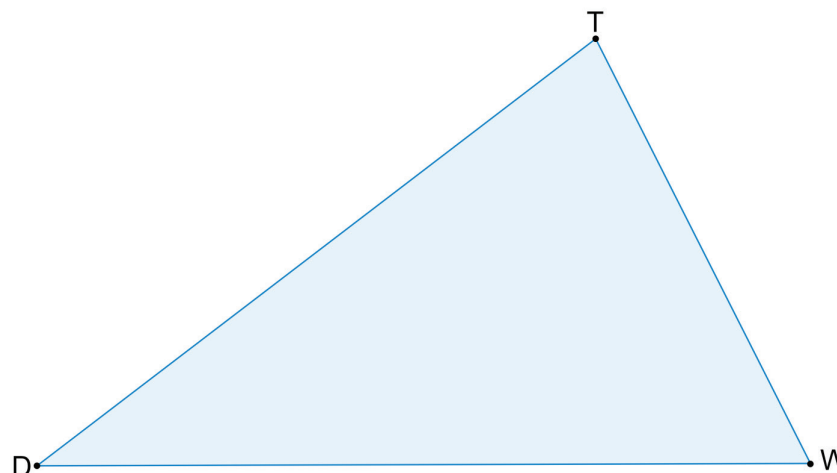
4. Review your conjecture with your partner and if necessary, make any changes.



A **circumscribed circle** is the smallest circle that includes a plane figure. If the figure is a polygon, then the circle must contain all of the vertices of the polygon. Not every polygon has a circumscribed circle, but all triangles and all regular polygons have circumscribed circles.

The **point of concurrency** is the point where three or more lines intersect.

5. Copy $\triangle DTW$ onto patty paper. Construct the perpendicular bisectors of each side. Name the point of concurrency of $\triangle DTW$ point R .



6. Discuss with your partner the relationship point R has with each of the vertices.

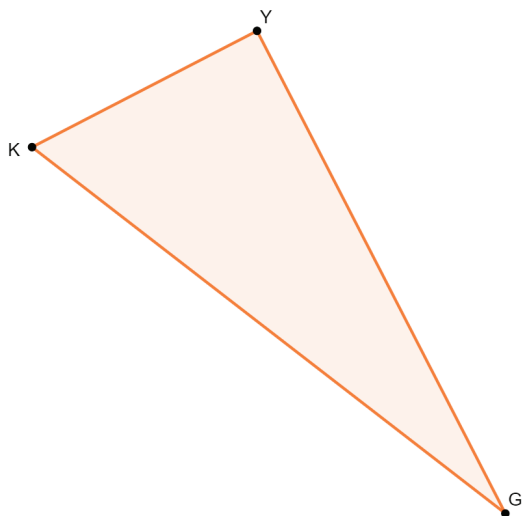
The point of concurrency of the perpendicular bisectors of a triangle (the center of a triangle's circumscribed circle) is the **circumcenter**.

7. Use the definition of a circumcenter to explain how the relationship point R has with each of the vertices ensures that a circle can be drawn to circumscribe the triangle.

8. Use a compass centered at point R to draw the circle that will pass through all three vertices.

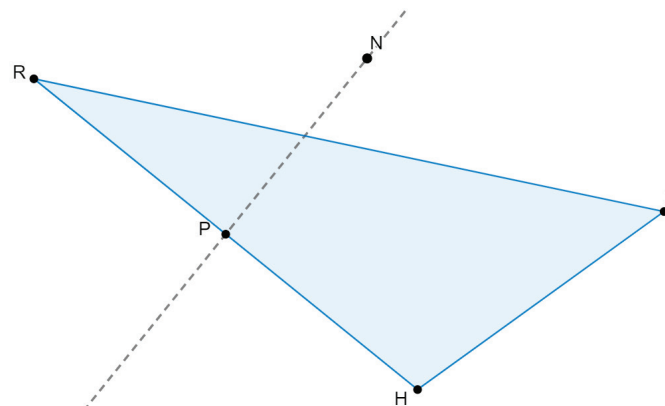
9. Review your conjecture. Does your conjecture hold true for newly constructed circumscribed Circle R ?

10. Copy $\triangle KYG$ onto another piece of paper. Then, use a compass and a straight edge to construct the circle that circumscribes $\triangle KYG$.



11. Describe how the circumscribed circle can be used to determine the type of triangle that $\triangle KYG$ is. Name a property and theorem from Unit 10 that can be used to verify this.

12. The dotted line is the perpendicular bisector of side $\overline{RH}$. The distance between points R and N is 4.35 units. If point N is the circumcenter, what is the distance between points H and N ? Explain or show your reasoning.

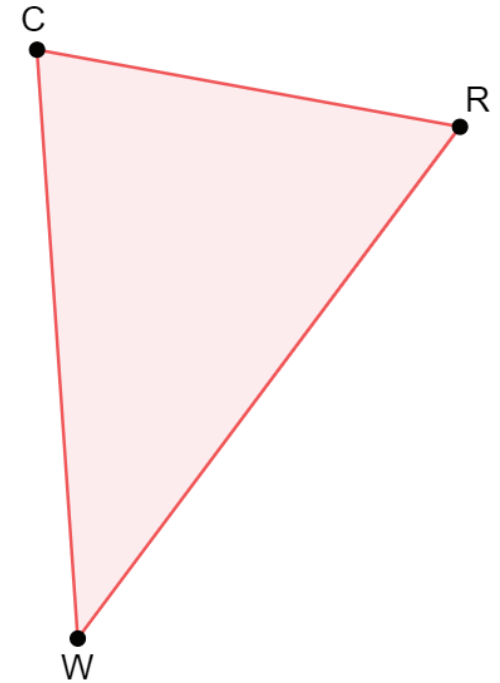


13. Emilio tells his partner, Talia, that in order to show that point N is the circumcenter, he only needs to construct one more perpendicular bisector. Talia challenged him to prove it. Describe how Emilio could justify that point N is the circumcenter by drawing only one more perpendicular bisector.



11.1.3 Wrap-Up: Ticket Out the Door

1. Describe how to construct Circle H , the circumscribed circle of $\triangle CRW$.



Unit 11, Lesson 2: Constructing the Incenter of a Triangle



Benchmark

MA.912.GR.5.3 Construct the inscribed and circumscribed circles of a triangle.

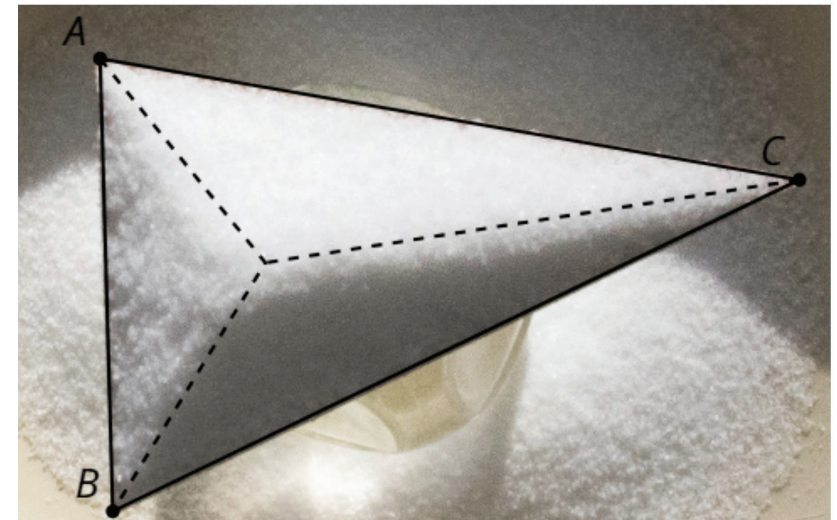


Learning Targets

- ☐ I can construct an inscribed circle in a triangle.
- ☐ I can locate the incenter of a triangle.



11.2.1 Warm-Up: Salt Pile



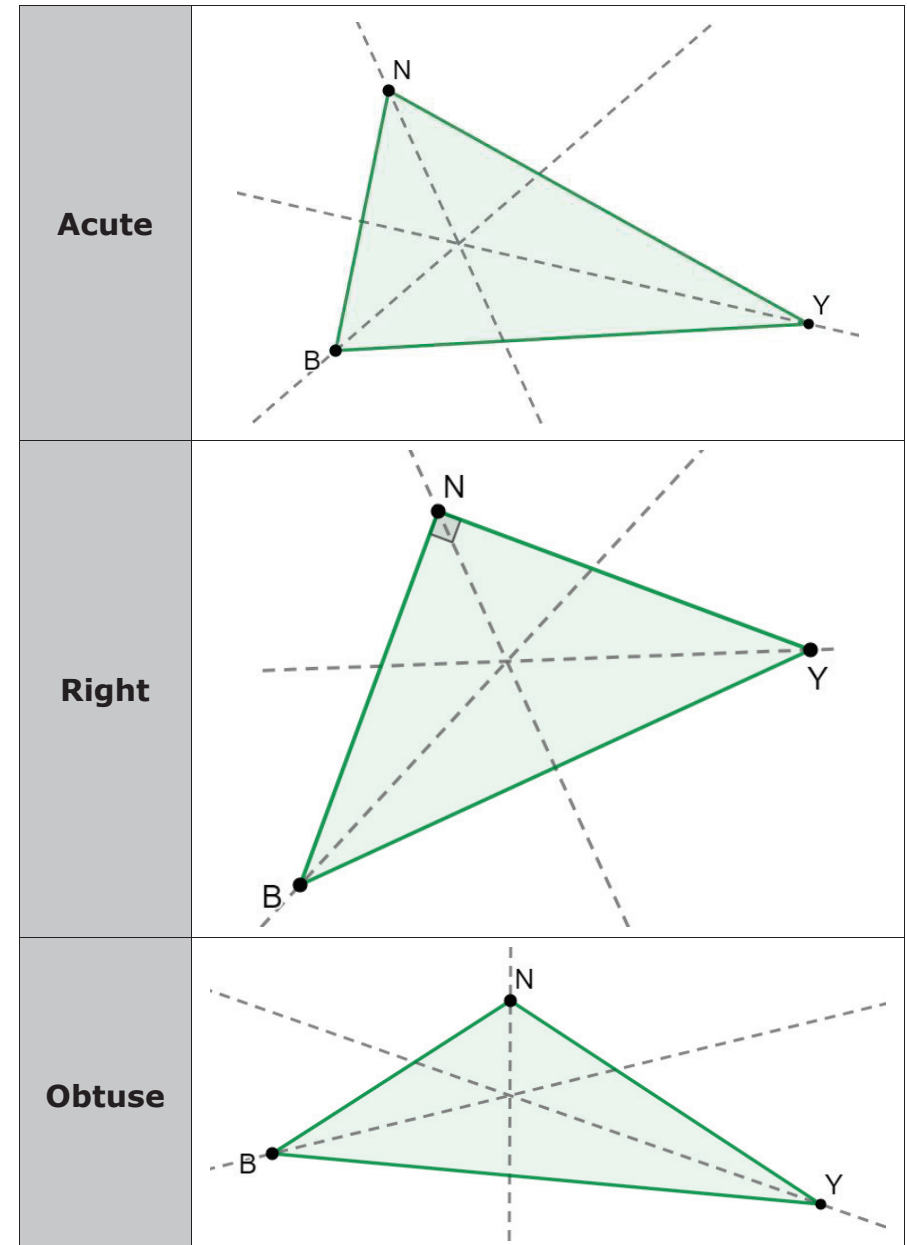
Salt is poured onto a triangle until it piles up. As the salt piles up, the pile reaches a maximum height. As the salt reaches a maximum height, new grains of salt fall off whichever side of the triangle is closest.

1. What do you wonder about the salt pile?
2. What do you notice about the salt pile?
3. What do you notice about the point where the salt reaches a maximum?



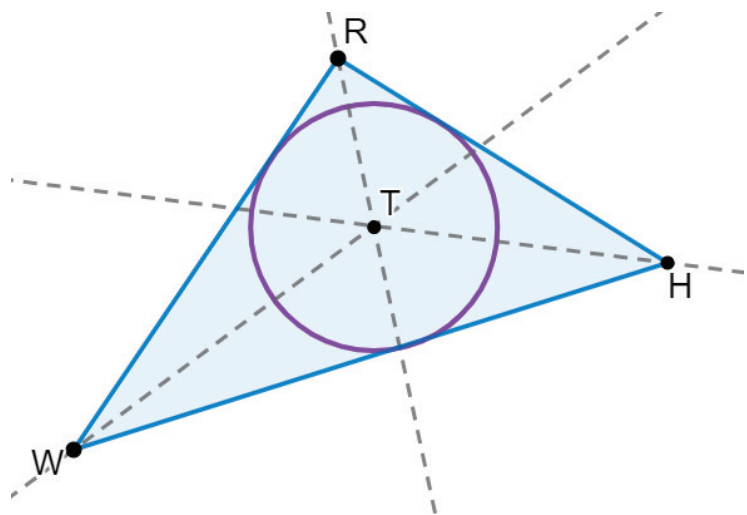
11.2.2 Exploration: Constructing an Inscribed Circle

1. $\triangle NYB$ is shown as three different types of triangles.
 - a. Label each point of concurrency as point R .



- b. What do you notice about the three different triangles and point R , the point of concurrency?

2. $\triangle RHW$ is shown where each angle is bisected. The three angle bisectors meet at point T .

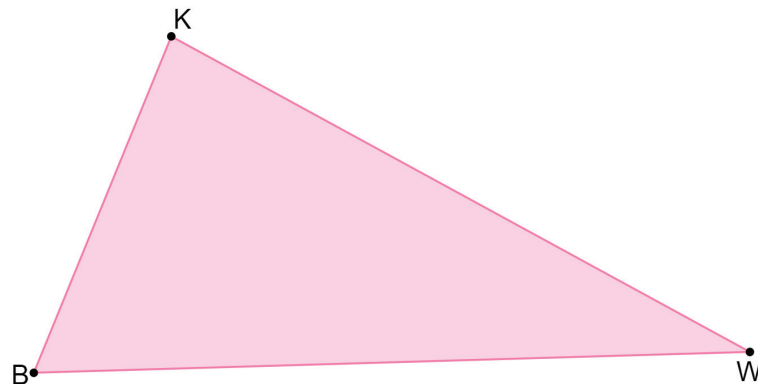


How is circle T related to $\triangle RHW$?

3. Use a compass to draw the inscribed circles for each type of $\triangle NYB$ in problem 1.

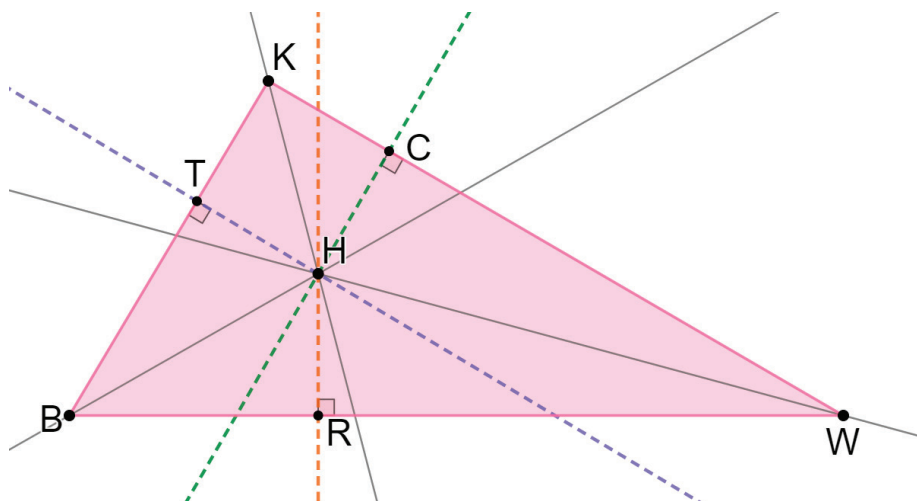
The point of concurrency of the angle bisectors of a triangle is the **incenter**.

4. Consider $\triangle KWB$.



- a. Copy $\triangle KWB$ onto patty paper and find the incenter of the triangle by constructing the angle bisectors. Label the incenter of $\triangle KWB$ as point H .
- b. Draw the inscribed circle for $\triangle KWB$ using point H , the incenter, as the center of the circle.

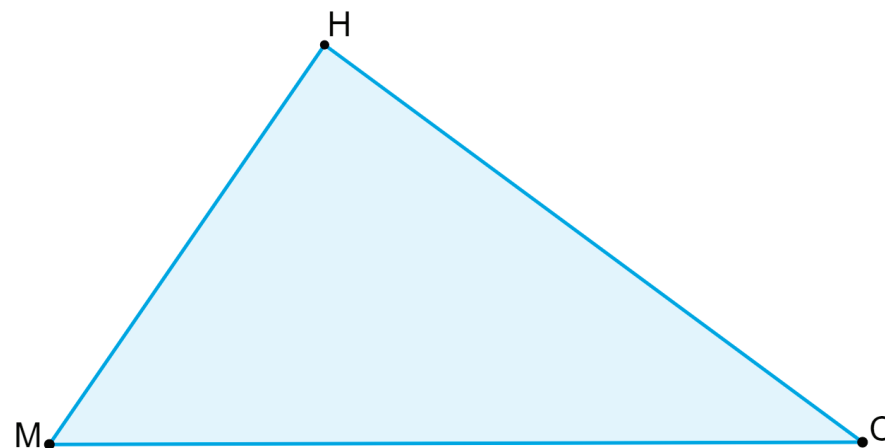
- c. Noah is concerned that when he draws the inscribed circle, he does not have the exact points on the triangle where the circle intersects the triangle. He drew three perpendicular lines from the center to each side.



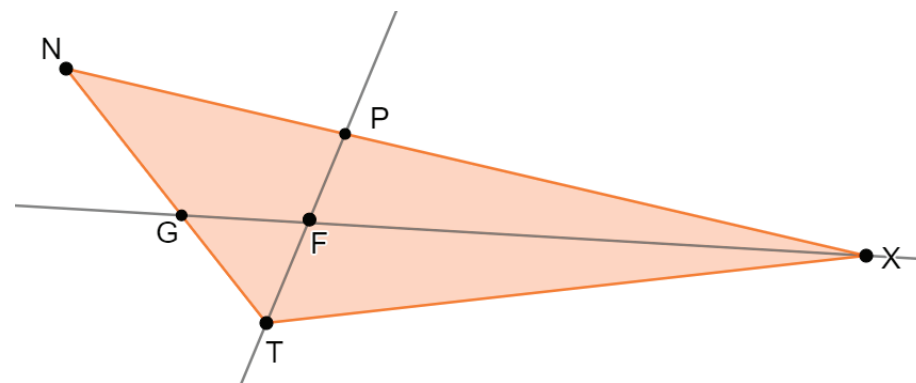
When Circle H is drawn on Noah's version of $\triangle KWB$, what relationship do $\overline{HR}$, $\overline{HT}$, and $\overline{HC}$ have to each other and to Circle H ?

- d. Draw Circle H that passes through points T , C , and R .

5. Copy $\triangle MHC$ onto another piece of paper. Then, use a compass and a straight edge to inscribe a circle in $\triangle MHC$.



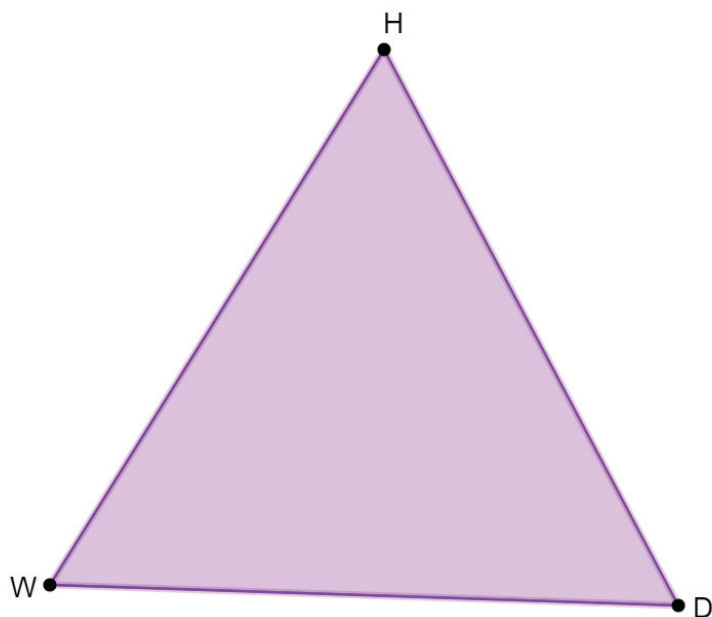
6. $\triangle NXT$ is shown where $\overline{PT}$ bisects $\angle NTX$ and $\overline{GX}$ bisects $\angle PXT$.



- a. Will the inscribed circle of $\triangle NXT$ pass through points P and G ?

- b. Justify why $\overline{PF}$ and $\overline{FG}$ are not radii of the inscribed Circle F.

7. Copy $\triangle HDW$ onto a piece of patty paper. Then, construct the incenter and circumcenter of $\triangle HDW$.



11.2.3 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on constructing a circle inscribed in a triangle?



Unit 11, Lesson H1: Inscribing Polygons in Circles



Benchmark

MA.912.GR.5.4 Construct a regular polygon inscribed in a circle. Regular polygons are limited to triangles, quadrilaterals, and hexagons.

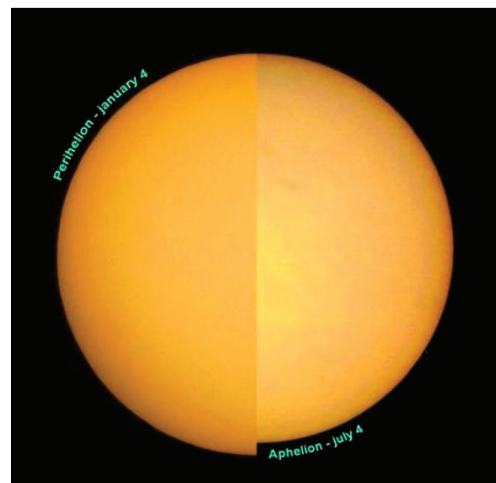


Learning Targets

- ☐ I can construct an equilateral triangle inscribed in a circle.
- ☐ I can construct a square inscribed in a circle.
- ☐ I can construct a regular hexagon inscribed in a circle.



11.H1.1 Warm-Up: Earth's Orbit



Since Earth's orbit is an ellipse, the Earth is slightly closer to the sun in January. The picture shows two photos taken by the SOHO satellite. The left side shows the sun on January 4, 2009 and the right side shows the sun on July 4, 2009.

A scientist measured the two images. She found that the diameter of the image for January was 72 millimeters and the image for July was 69 millimeters.

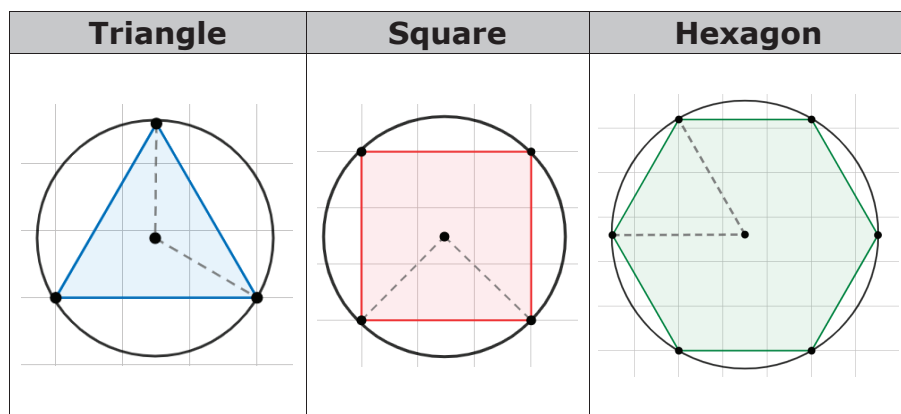
1. By what percentage did the diameter of the Sun change between January and July compared to its average diameter of January and July?
2. If the average distance to the Sun from Earth is 149,600,000 kilometers, how much closer is Earth to the Sun in January compared to July?



11.H1.2 Activity: Regular Polygons Inscribed in Circles

Work with your partner to answer each question. Use the larger copies of each polygon provided by your teacher.

1. Use a compass to draw a circumscribed circle for each polygon.
2. Mark the center of the circle that was drawn on the polygon. Draw a central angle that passes through two subsequent vertices, as shown.



3. Use a protractor to determine the measure of each central angle.

	Triangle	Square	Hexagon
Measure of Central Angle			

4. Make a conjecture about how to construct each regular polygon inscribed in a circle using a compass and a straight edge.

	Construction
Triangle	
Square	
Hexagon	

5. Ask another pair of partners for the conjectures they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first three columns of the table.* Have the person initial next to your summary stating that your summary was accurate.

	Other Partner Pair's Conjecture	Other Partner Pair's Initials
Triangle		
Square		
Hexagon		

6. Review your conjectures with your partner and if necessary, make any changes.



11.H1.3 Activity: Card Sort

Sort the sets for each construction. Then, write the steps in the table.

	Construction Steps
Triangle Method 1	
Square Method 1	

	Construction Steps
Hexagon Method 1	
Triangle Method 2	

	Construction Steps
Square Method 2	
Hexagon Method 2	



11.H1.4 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about inscribing polygons in circles.

Unit 11, Lesson 3: Solving Problems Involving Triangles Inscribed in Circles



Benchmark

MA.912.GR.6.3 Solve mathematical problems involving triangles and quadrilaterals inscribed in a circle.



Learning Target

- ☐ I can solve mathematical problems involving triangles inscribed in a circle.



11.3.1 Warm-Up: Patterns



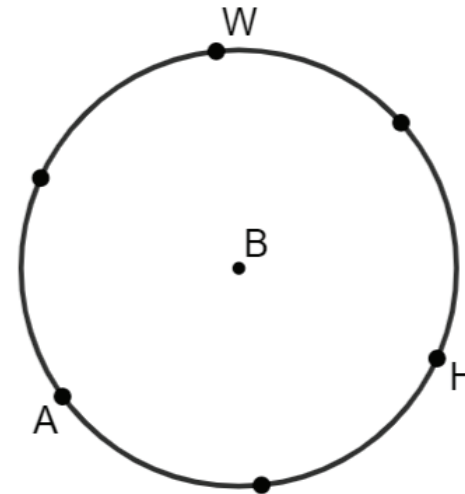
1. What is the first question that comes to your mind?
2. Describe the pattern used in the image.



11.3.2 Exploration: Triangles Inscribed in Circles

Work with your partner to complete the following.

Elijah is constructing a triangle inscribed in Circle B and stopped after he created six points on Circle B that are the same distance from each other.



1. Using a straightedge, draw a line from point A to point H .
2. Using a straightedge, draw a line from point H to point W .
3. In relation to the circle, what type of angle is $\angle AHW$?

4. Explain the relationship between $m\angle AHW$ and $m\widehat{WA}$.

5. Using a straightedge, draw a line from point A to point W .

6. Use a ruler or patty paper to verify that $\triangle AHW$ is an equilateral triangle.

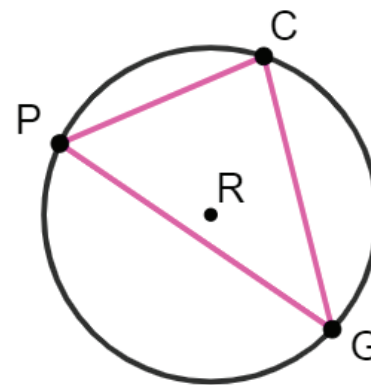
7. What is the $m\angle AHW$?

8. What is the $m\widehat{WA}$?



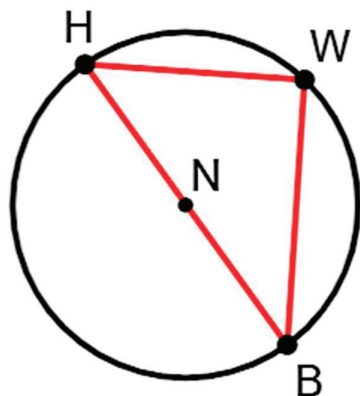
11.3.3 Guided Practice: Triangles Inscribed in Circles

1. Circle R is shown. Points P , C , and G are on circle R and create $\triangle PCG$. $m\widehat{PG} = 162^\circ$, $m\angle CPG = 58^\circ$, and $m\angle PCG = (7x - 17)^\circ$.



- What is the $m\angle PCG$?
- Find the value of x .
- What is the $m\angle PGC$?

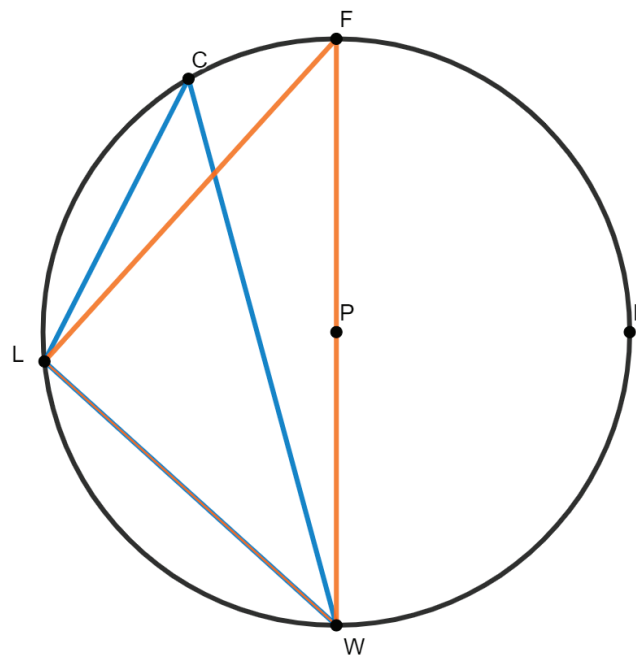
2. Circle N has points H , W , and B on the circle and diameter $\overline{HB}$. $\triangle HWB$ is inscribed in circle N . $m\widehat{BW} = 102^\circ$ and $m\angle HBW = (4x - 10)^\circ$.



a. What is the $m\angle HBW$?

b. Find the value of x .

3. $\triangle CLW$ and $\triangle FLW$ are inscribed in Circle P , where $\overline{FW}$ is the diameter of Circle P . $m\widehat{FC} = (2x - 6)^\circ$, $m\widehat{LC} = (4x - 6)^\circ$, and $m\widehat{LW} = (3x + 30)^\circ$.



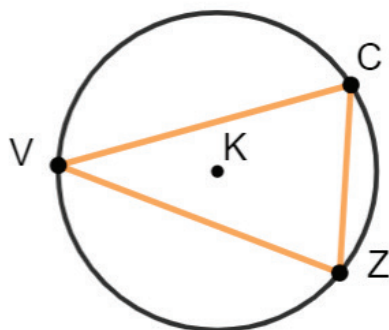
Complete the table.

Angle Measures	Angle Measures
$m\angle LCW =$	$m\angle LFW =$
$m\angle CWL =$	$m\angle FWC =$
$m\angle FLC =$	$m\angle FLW =$



11.3.4 Your Turn!

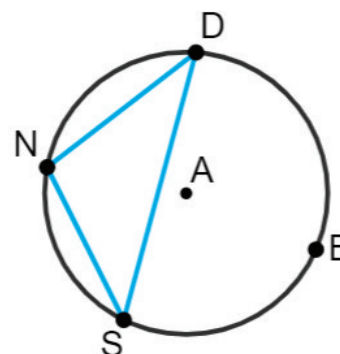
1. Circle K has points V , C , and Z on the circle, and $\triangle VCZ$ is inscribed in circle K . $m\widehat{VC} = (9x + 23)^\circ$, $m\angle CVZ = 36^\circ$, and $m\angle VZC = (7x - 21)^\circ$.



a. What is the value of x ?

b. What is the $m\angle VCZ$?

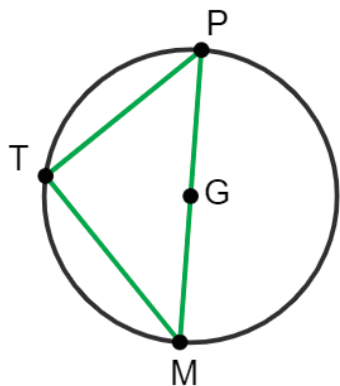
2. Circle A has points N , D , B , and S on the circle, and $\triangle SDN$ is inscribed in Circle A . $m\widehat{ND} = (7x - 7)^\circ$, $m\widehat{DBS} = (23y - 5)^\circ$, $m\angle NSD = (5x - 23)^\circ$, and $m\angle DNS = (8y + 29)^\circ$.



Complete the table.

Angle Measures	Arc Measures
$m\angle NSD =$	$m\widehat{ND} =$
$m\angle SND =$	$m\widehat{DBS} =$
$m\angle NDS =$	$m\widehat{NS} =$

3. Circle G has diameter $\overline{MP}$ and $\triangle MTP$ inscribed in the circle. $m\angle TMP = (6x - 8)^\circ$ and $m\angle TPM = (4x + 13)^\circ$.



- Find the value of x .
- What is the $m\angle TMP$?
- What is the $m\angle TPM$?



11.3.5 Wrap-Up: Cool Down

- Write a summary of what you learned about solving problems involving triangles inscribed in circles.

Unit 11, Lesson 4: Solving Problems Involving Quadrilaterals Inscribed in Circles



Benchmark

MA.912.GR.6.3 Solve mathematical problems involving triangles and quadrilaterals inscribed in a circle.



Learning Target

- ☐ I can solve mathematical problems involving quadrilaterals inscribed in a circle.



11.4.1 Warm-Up: Notice and Wonder

Machu Picchu is an Inca citadel in southern Peru. Machu Picchu was built in the classical Inca style with polished dry-stone walls and consists of more than 150 buildings. Although many of the stones that were used to build the city were more than 50 pounds, it is believed that no wheels were used to transport these rocks up the mountain.

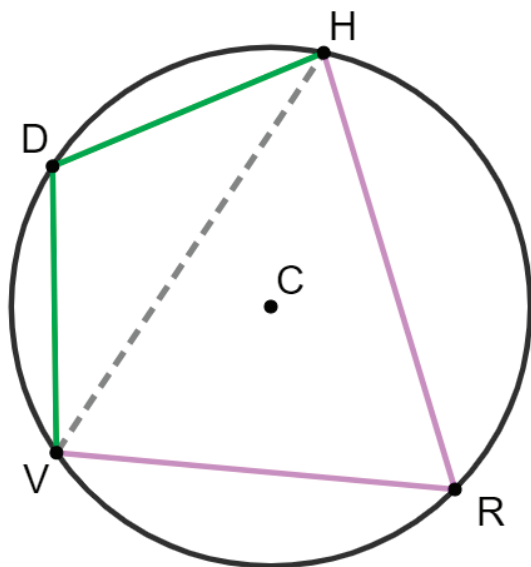


1. What do you notice?
2. What do you wonder?



11.4.2 Exploration: Quadrilaterals Inscribed in Circles

1. Circle C is shown, where $\overline{DV} \cong \overline{DH}$. $m\widehat{RV} = 100^\circ$, $m\widehat{DV} = 68^\circ$, $m\widehat{HDV} = 136^\circ$, and $m\widehat{HRV} = 224^\circ$.



- a. Complete the table.

Angle Measures	Angle Measures
$m\angle RHV =$	$m\angle HDV =$
$m\angle DHV =$	$m\angle DVH =$
$m\angle HRV =$	$m\angle HVR =$

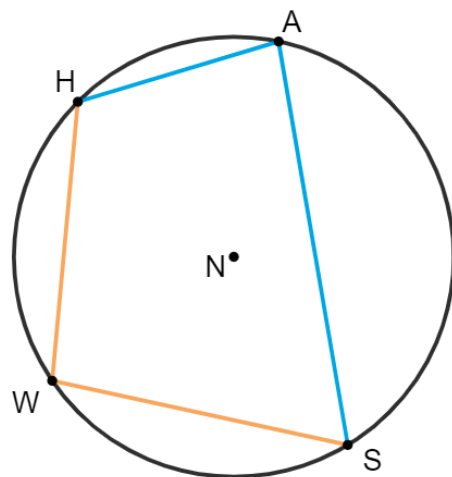
- b. Work with your partner to make a conjecture about the relationships of the angles in $DHRV$.

- c. Ask two classmates for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Reason	Initials

- d. Review your conjecture with your partner. If necessary, make any changes.

2. Circle N has points H , A , S , and W on the circle. Quadrilateral $HASW$ is inscribed in the circle. $m\widehat{HA} = 59^\circ$, $m\widehat{HWS} = 166^\circ$ and $m\widehat{WHA} = 136^\circ$.



Use your conjecture to find the measures.

Angle Measures

$$m\angle WHA =$$

$$m\angle HAS =$$

$$m\angle ASW =$$

$$m\angle SWH =$$

Arc Measures

$$m\widehat{SA} =$$

$$m\widehat{SW} =$$

$$m\widehat{HW} =$$



11.4.3 Guided Practice: Quadrilaterals Inscribed in Circles

1. Complete the statement.

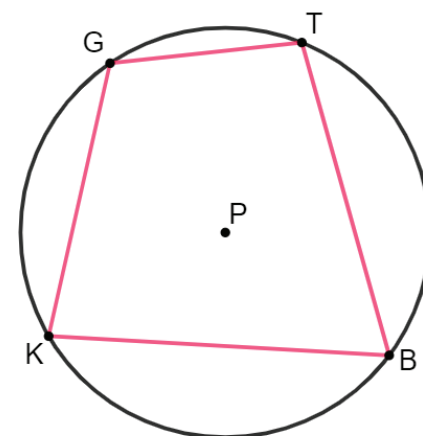
If a quadrilateral is inscribed in a circle,

- ☐ adjacent
- ☐ opposite

angles are

- ☐ congruent.
- ☐ complementary.
- ☐ supplementary.

2. Circle P has $GTBK$ inscribed in the circle. $m\widehat{TBK} = 206^\circ$ and $m\angle GKB = 84^\circ$.

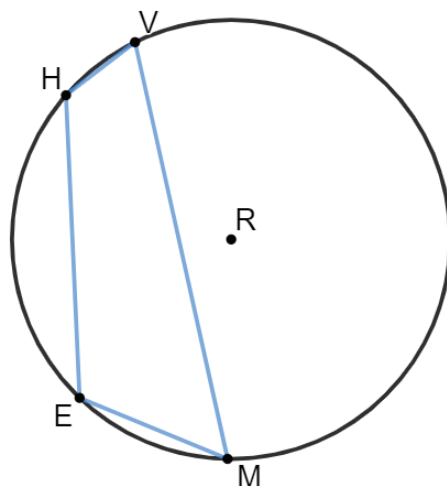


- a. What is the $m\angle TBK$?

- b. What is the $m\angle GTB$?

3. Circle R has $VMEH$ inscribed in the circle.
 $m\angle HEM = (2x - 7)^\circ$,
 $m\angle EHV = (2x + 3)^\circ$, and
 $m\angle HVM = (x + 4)^\circ$.

a. Find the value of x .



b. Complete the table.

$m\angle HVM =$
$m\angle HEM =$
$m\angle EHV =$
$m\angle EMV =$



11.4.4 Your Turn!

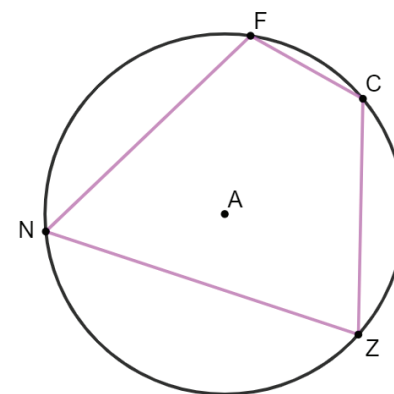
1. Circle A has $NFCZ$ inscribed in the circle.
 $m\angle NFC = (6x + 32)^\circ$,
 $m\angle FCZ = (6y - 8)^\circ$,
 $m\angle NZC = 73^\circ$, and
 $m\angle FNZ = (3y - 1)^\circ$.

a. Find the value of x .

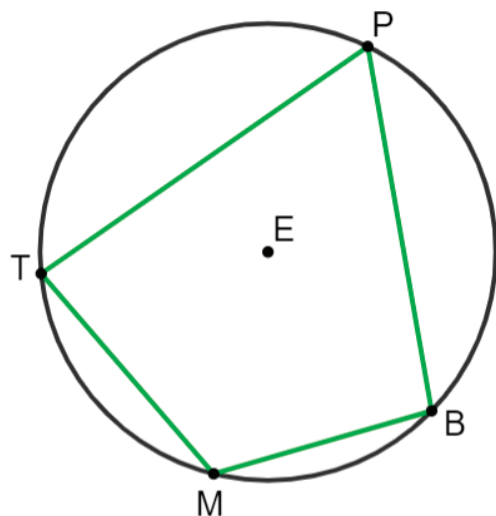
b. What is the $m\angle NFC$?

c. Find the value of y .

d. What is the $m\angle FCZ$?



2. Circle E has $TPBM$ inscribed in the circle. $m\widehat{TM} = 71^\circ$, $m\widehat{MB} = 61^\circ$, and $m\angle TMB = (12x - 3)^\circ$.



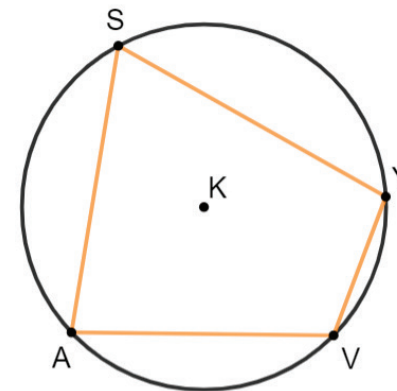
- Find the value of x .
- What is the $m\angle TMB$?



11.4.5 Wrap-Up: Error Analysis

1. $SYVA$ is inscribed in Circle K , where $m\angle ASY = (3x + 23)^\circ$ and $m\angle AVY = (6x + 13)^\circ$.

Anastasia is finding the value of x . Her work is shown.



Anastasia's Work

$$\begin{aligned} 3x + 23 &= 6x + 13 \\ 3x &= 10 \\ x &= \frac{10}{3} \end{aligned}$$

- Write a note to Anastasia explaining her mistake.
- Find the value of x .

Unit 11, Lesson 5: Interior and Exterior Angles of Triangles



Benchmark

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.



Learning Targets

- ☐ I can show that the sum of the interior angles of a triangle is 180 degrees.
- ☐ I can show that the sum of the exterior angles of a triangle is 360 degrees.
- ☐ I can solve problems involving postulates, relationships, and theorems of triangles.



11.5.1 Warm-Up: Superhero

1. If geometry had a superpower, it would be ...

because ...

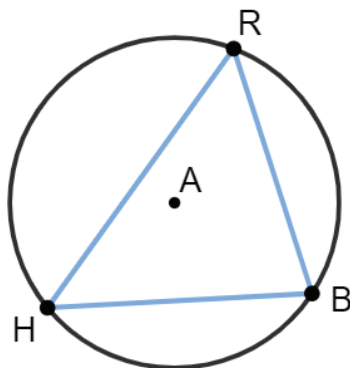
2. Geometry's villain would be ...

because ...



11.5.2 Exploration: Interior Angles of Triangles

1. Circle A is shown, where $m\widehat{RB} = 102^\circ$, $m\widehat{BH} = 108^\circ$, and $m\widehat{HR} = 150^\circ$.

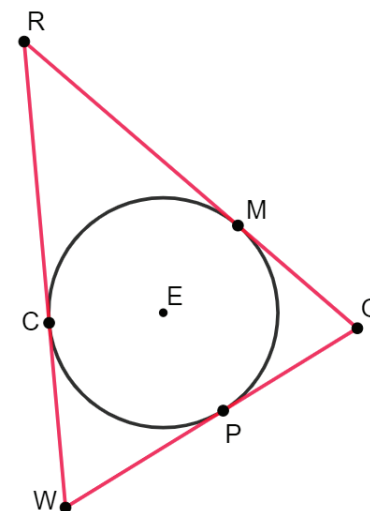


- a. Complete the table.

Angle Measures
$m\angle RHB =$
$m\angle RBH =$
$m\angle HRB =$

- b. What is the sum of the angles in $\triangle RHB$?

2. Circle E is shown where points P , C , and M are on the circle. $\overline{RG}$ and $\overline{RW}$ are tangent to Circle E and intersect at point R outside the circle. $\overline{GR}$ and $\overline{GW}$ are tangent to Circle E and intersect at point G outside the circle. $\overline{WR}$ and $\overline{WP}$ are tangent to Circle E and intersect at point W outside the circle. $m\widehat{CM} = 136^\circ$, $m\widehat{MP} = 108^\circ$, and $m\widehat{PC} = 116^\circ$.



- a. Complete the table.

Angle Measures
$m\angle WRG =$
$m\angle RGW =$
$m\angle GWR =$

- b. What is the sum of the angles in $\triangle RGP$?
- c. Write a conjecture about the sum of the interior angles of a triangle.

- d. Ask two classmates for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Reason	Initials

- e. Review your conjecture with your partner. If necessary, make any changes.



Triangle Sum Theorem

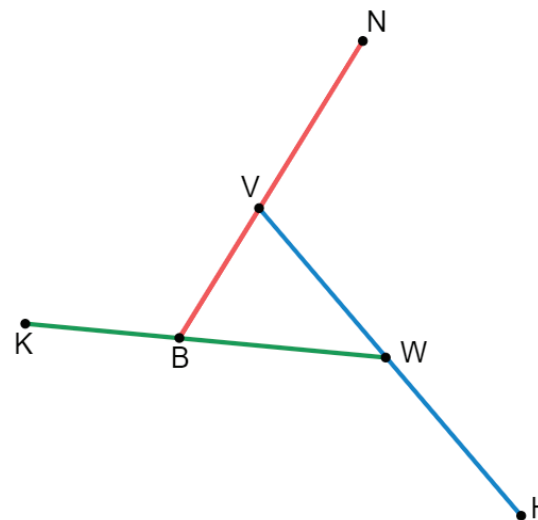
The sum of the measures of the interior angles of a triangle is 180° .



11.5.3 Activity: Exterior Angles of Triangles

Work with your partner to complete the following.

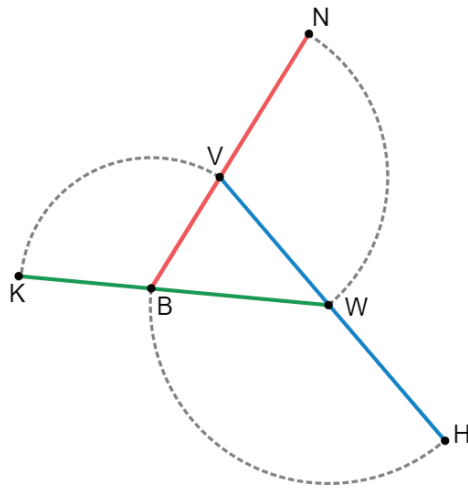
1. On a separate sheet of paper, draw three line segments, $\overline{BN}$, $\overline{VH}$, and $\overline{KW}$, to make $\triangle VBW$ as shown.



An **exterior angle** is an angle created by extending one leg of a triangle.

- List the exterior angles of $\triangle VBW$.
- Estimate the sum of the exterior angles of $\triangle VBW$.

2. Use a compass to draw arc $\widehat{BH}$ using point W as the center, $\widehat{KV}$ using point B as the center, and $\widehat{NW}$ using point V as the center, as shown.



- Trace the figure above onto patty paper and cut out each sector. Rearrange the sectors to determine the sum of the exterior angles.
- What do you notice about the sum of the exterior angles of the triangle?
- Write a conjecture about the sum of the exterior angles of a triangle.

- Ask two classmates for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

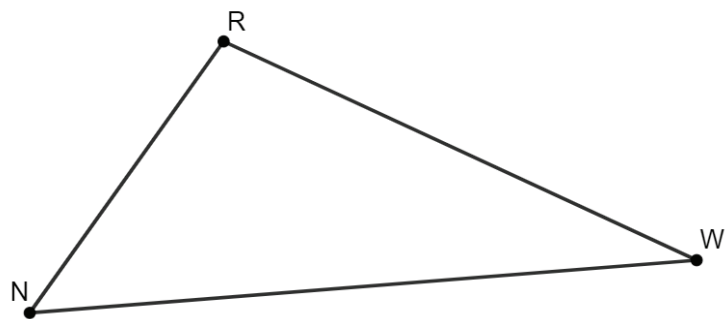
Conjecture	My Summary of Their Reason	Initials

- Review your conjecture with your partner. If necessary, make any changes.



11.5.4 Guided Practice: Interior and Exterior Angles of Triangles

1. $\triangle RWN$ is shown where $m\angle NRW = (14x + 3)^\circ$, $m\angle RWN = (\frac{9}{2}x - \frac{5}{2})^\circ$, and $m\angle WNR = (9x - 13)^\circ$.

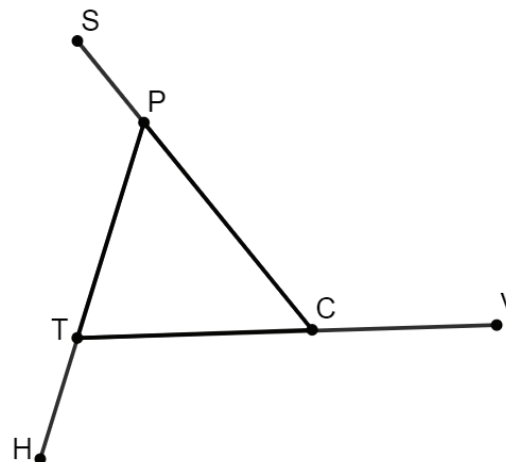


- a. Find the value of x .

- b. Complete the table.

$m\angle NRW =$	$m\angle RWN =$	$m\angle WNR =$
-----------------	-----------------	-----------------

2. $\triangle PCT$ is shown, where $\angle SPT$, $\angle PCV$, and $\angle HTC$ are exterior angles. $m\angle PCT = 52^\circ$, $m\angle SPT = (11x - 20)^\circ$, and $m\angle HTC = (8x + 5)^\circ$.



- a. Find the value of x .

- b. What is the measure of $\angle SPT$?

- c. What is the measure of $\angle HTC$?



11.5.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.



Unit 11, Lesson 6: Triangle Inequality Theorem



Benchmarks

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can determine if a set of triangle side lengths can be used to construct a triangle.
- ☐ I can use the triangle inequality theorem to prove relationships in triangles.



11.6.1 Warm-Up: Number Riddle

1. Use any of the four operations (+, -, $\times$, $\div$) with the numbers shown to produce a result of 36.

7, 0, 2, 3, 5



11.6.2 Exploration: Triangle Inequality Theorem

1. With your partner, use a ruler to determine if the given measurements can be used to create a triangle. Then, complete the table.

Measurements in centimeters (cm)	Can it create a triangle?	Explanation
3 cm, 6 cm, and 8 cm	☐ Yes ☐ No	
3 cm, 4 cm, and 10 cm	☐ Yes ☐ No	
8 cm, 11 cm, and 15 cm	☐ Yes ☐ No	
2 cm, 5 cm, and 8 cm	☐ Yes ☐ No	
4 cm, 5 cm, and 11 cm	☐ Yes ☐ No	

2. What similarities did you notice about the measurements that could create a triangle?
3. What similarities did you notice about the measurements that could not create a triangle?

4. Make a conjecture about how to determine if side lengths can be used to construct a triangle if you cannot draw it.

5. Ask two other pairs of partners for the conjectures they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

My Summary of the Conjecture	Initials

6. Review your conjecture with your partner and if necessary, make any changes.

7. Use your conjecture to determine if 3 cm, 4 cm, and 7 cm will create a triangle.

8. Check your answer by constructing the triangle with a ruler.

9. What did you notice about the measurements 3 cm, 4 cm, and 7 cm?

10. Adjust your conjecture if needed.



Triangle Inequality Theorem

The sum of the lengths of any two sides of a triangle is greater than the length of the third side.



If three segments have lengths such that the sum of any two is greater than the third, then a triangle can be constructed from the segments.



11.6.3 Try It: Triangle Inequality Theorem

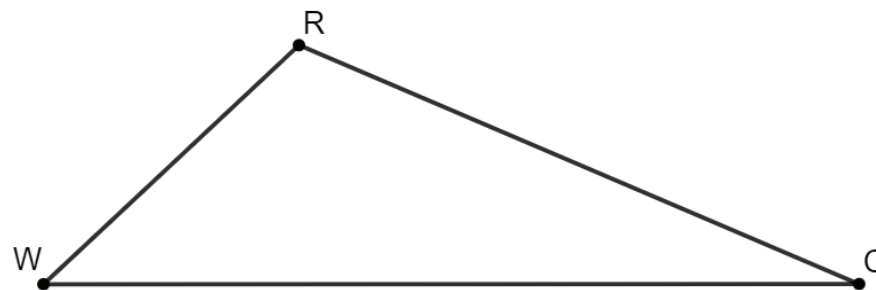
- Complete the table.

Side Lengths	Work	Can it create a triangle?
57, 16, and 75		☐ Yes ☐ No
33, 12, and 25		☐ Yes ☐ No
21, 5, and 26		☐ Yes ☐ No
61, 100, and 61		☐ Yes ☐ No
48, 24, and 21		☐ Yes ☐ No



11.6.4 Exploration: Relating Angle Measures and Sides of Triangles

$\triangle RCW$ is shown, where $WC = 24$, $CR = 18$, and $RW = 10$.



- Use a protractor to find the angle measures. Record your answers in the table.

Angle Measures
$m\angle CWR =$
$m\angle WRC =$
$m\angle RCW =$

- List the side lengths in order from longest to shortest.
- List the angles in order from largest to smallest.

4. What do you notice about the size of the angle compared to the length of the opposite side?



11.6.5 Guided Instruction: Proving Inequality in a Triangle



If two **sides** of a triangle are unequal in measure, then the angle opposite the longer side has the greater measure.



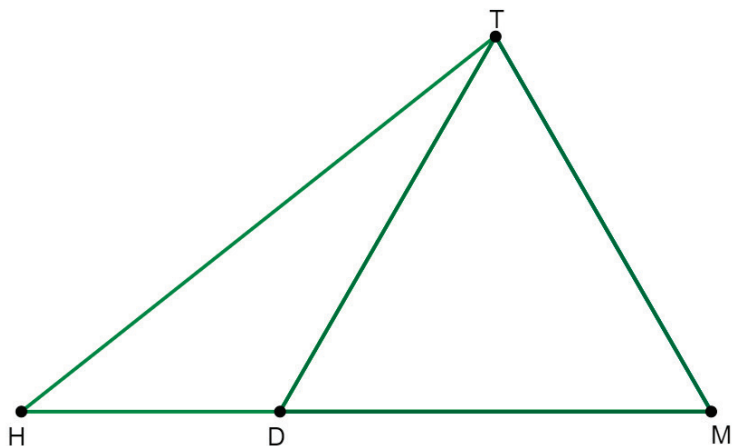
If two **angles** of a triangle are unequal in measure, then the side opposite the angle with the greater measure is the longer side.



Whole Greater Than Parts Theorem

If $a > 0$, $b > 0$, and $a + b = c$, then $c > a$ and $c > b$.

1. $\triangle THM$ is shown, with $\overline{TD}$ creating $\triangle HTD$ and $\triangle DTM$, where $\triangle DTM$ is an equilateral triangle.



Given: $\triangle DTM$ is equilateral; $m\angle DTH = 22^\circ$

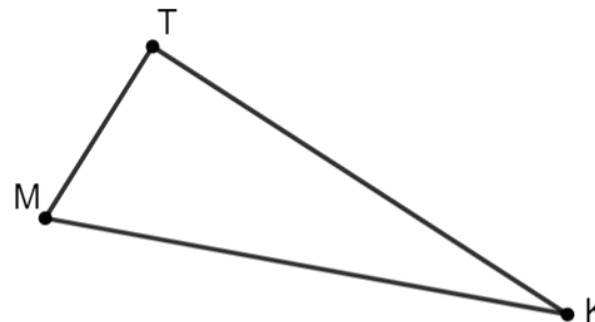
Prove: $HM > HT$

Statement	Reason
1. $\triangle DTM$ is equilateral; $m\angle DTH = 22^\circ$	1. Given
2.	2. Property of equilateral triangles
3. $m\angle DTM + m\angle DTH = m\angle MTH$	3.
4.	4. Substitution Property of Equality
5. $m\angle MTH > m\angle TMD$	5. Whole greater than parts theorem
6. $HM > HT$	6.



11.6.6 Try It: Triangle's Angle Measures

1. $\triangle TKM$ is shown, where $KT = 32$, $MK = 34$, and $TM = 13$. List the angles in order from least to greatest.





11.6.7 Wrap-Up: Cool Down

1. List three side lengths that would not make a triangle. Explain how you know those sides would not make a triangle.
2. List three side lengths that would make a triangle. Explain how you know those sides would make a triangle.



Unit 11, Lesson 7: Centroid of a Triangle



Benchmarks

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.

MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.



Learning Targets

- ☐ I can find the centroid of a triangle.
- ☐ I can use the relationship of the distance a centroid is from a vertex to the midpoint of the opposite side to solve problems.



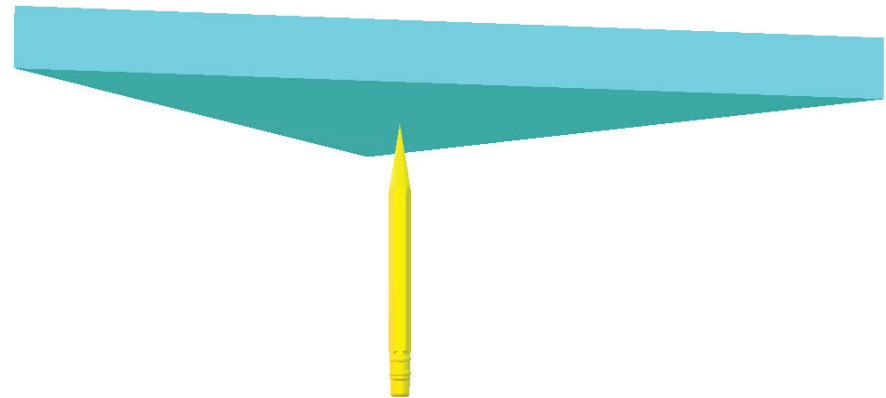
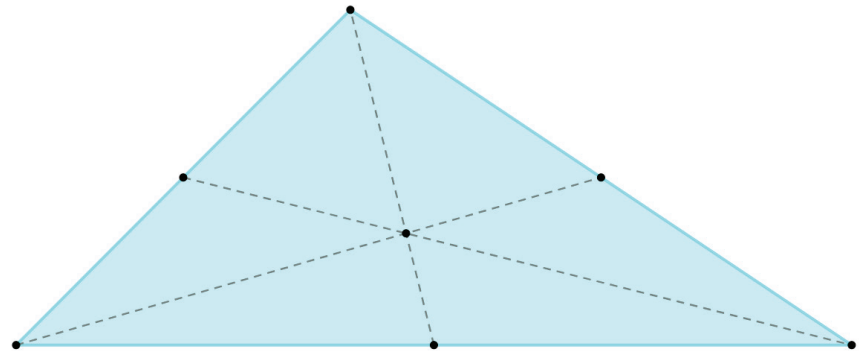
11.7.1 Warm-Up: Bell Work

1. Given $\triangle BMT$, where $MB = 7$, $MT = 12$, and $BT = 9$, list the angles in order from least to greatest.



11.7.2 Exploration: Centroid of a Triangle

1. All polygons have a balancing point. The diagrams show the balancing point of a triangle.

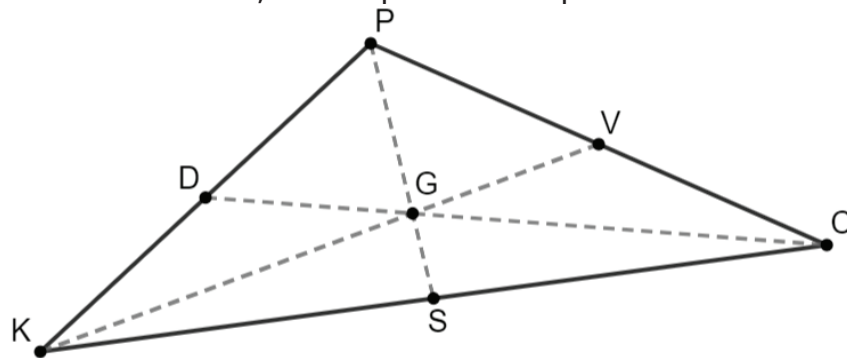


- a. What do you notice?

- b. Discuss with your partner how you think the balance point was found. Summarize your discussion.

A median of a triangle is a segment that joins any vertex of the triangle to the midpoint of the opposite side.

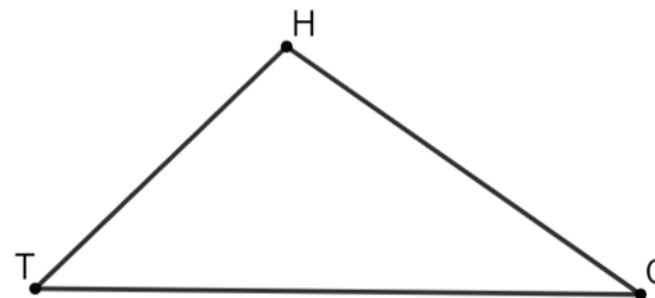
2. $\triangle KPC$ is shown, where point G is a point of concurrency.



Discuss with your partner how to find the midpoints of the triangle's sides using each method. Complete the table.

Patty Paper	Compass

3. $\triangle HCT$ is shown.

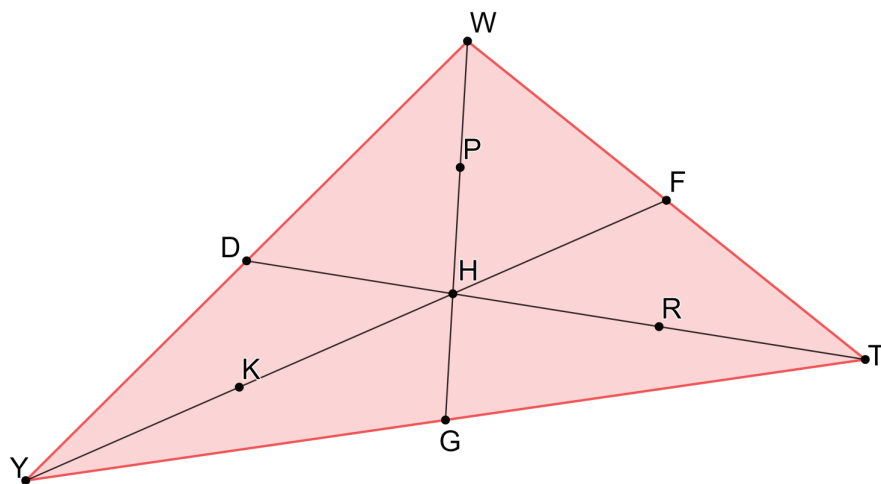


- a. Use the steps to construct the centroid of $\triangle HCT$.

Step 1: Trace $\triangle HCT$ on patty paper.
Step 2: Construct the perpendicular bisector of $\overline{HT}$ to find its midpoint, M .
Step 3: Draw the median from M to the opposite vertex, $\angle HCT$.
Step 4: Construct the perpendicular bisector of $\overline{HC}$ to find its midpoint, R .
Step 5: Draw the median from R to the opposite vertex, $\angle HCT$.
Step 6: Construct the perpendicular bisector of $\overline{TC}$ to find its midpoint, X .
Step 7: Draw the median from midpoint X to the opposite vertex, $\angle THC$.
Step 8: Draw point W , the point of concurrency, where medians $\overline{RT}$, $\overline{XH}$, and $\overline{MC}$ intersect.

- b. What do you notice about the location of point W in $\triangle HCT$?

4. $\triangle YWT$ is shown, with medians $\overline{WG}$, $\overline{TD}$, and $\overline{YF}$. Point P is the midpoint of $\overline{WH}$, point R is the midpoint of $\overline{TH}$, and Point K is the midpoint of $\overline{YH}$.



- a. Copy the first segment using patty paper. Use the copy to determine the relationship the first segment and the point have with the second segment. Complete the table.

First Segment	Point	Second Segment	Relationship
$\overline{DH}$	R	$\overline{TH}$	
$\overline{FH}$	K	$\overline{YH}$	
$\overline{GH}$	P	$\overline{WH}$	

The **centroid of a triangle** is the intersection of the three medians of the triangle (each median connecting a vertex with the midpoint of the opposite side).

- b. Write a conjecture about the distance a centroid is from the vertex and the distance a centroid is from the midpoint of the opposite side.



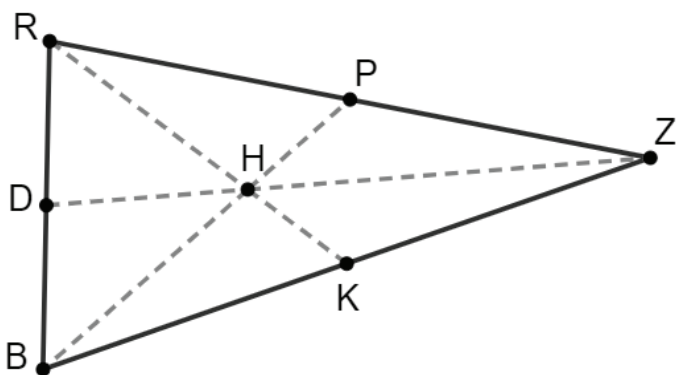
11.7.3 Guided Instruction: Centroid of a Triangle



The length from the vertex to the centroid is twice the length from the centroid to the midpoint of the opposite side, which yields a ratio of 2:1.

1. $\triangle RZB$ is shown, where point H is the centroid of the triangle.

$DH = 11.3$, $RK = 20.55$, and $BH = 15.2$.

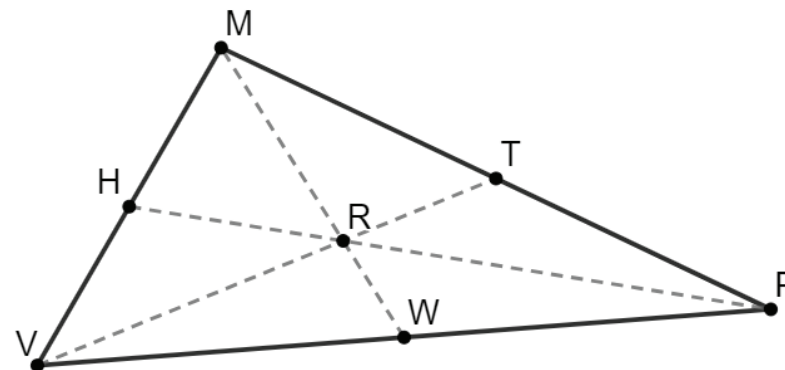


- a. Complete the table.

$HZ =$	$DZ =$
$RH =$	$HK =$
$HP =$	$BP =$

- b. If $BK = 6x + 3$ and $KZ = 10x - 7$, what is the length of $\overline{BZ}$?

2. Triangle MPV is shown, where point R is the centroid of the triangle. $MR = 5x - 9$, $RP = 3w + 7$, $RV = 12y - 9$, $RW = 1.75x$, $RT = 4y + 2$, and $HR = w + 9$.



- a. Find the value of each variable.

$x =$	$y =$	$w =$
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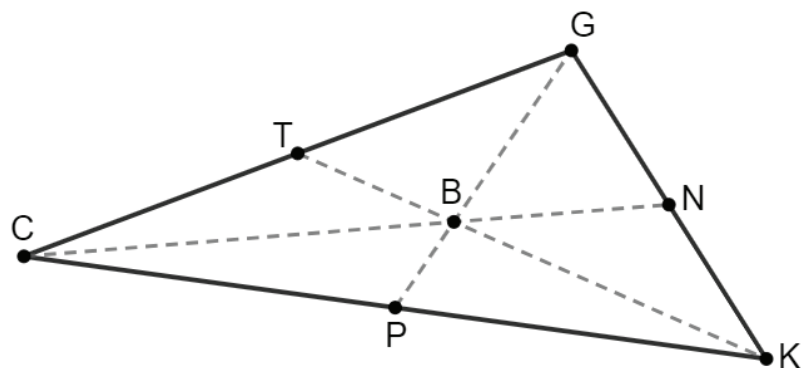
- b. Complete the table.

$HR =$	$RP =$	$HP =$
$RT =$	$VR =$	$VT =$
$RW =$	$MR =$	$MW =$



11.7.4 Your Turn!

1. $\triangle CGK$ is shown, where point B is the centroid of the triangle. $TB = \frac{1}{2}x - 4$, $PB = 4.4$, $CB = y + 6$, $BK = 2x - 31$, and $BN = \frac{1}{3}y + 5$.



a. What is the value of x ?

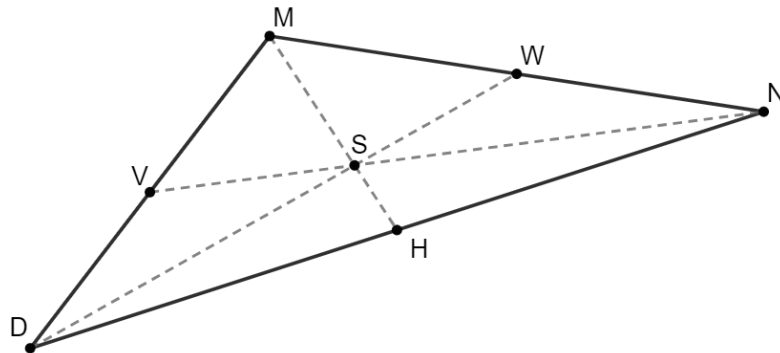
b. What is the value of y ?

- c. Complete the table.

$TB =$	$BK =$
$CB =$	$CN =$
$BG =$	$PG =$

- d. If $CT = 4k - 10$ and $TG = 2k + 1$, what is the length of $\overline{CG}$?

2. $\triangle DMN$ is shown, where point S is the centroid.



- a. Leonardo is finding the value of x using the information $VS = \frac{1}{2}x + 1$ and $SN = 2x - 6$. His work is shown.

Leonardo's Work

$$\begin{aligned} 2(2x - 6) &= \frac{1}{2}x + 1 \\ 4x - 12 &= \frac{1}{2}x + 1 \\ \frac{7}{2}x &= 13 \\ x &= \frac{26}{7} \end{aligned}$$

- b. Write a note to Leonardo explaining his mistake.

- c. What is the value of x ?



11.7.5 Wrap-Up: Cool Down

1. Write a short note summarizing today's lesson to a friend who missed class.

Unit 11, Lesson 8: Triangle Midsegment Theorem



Benchmarks

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can discover the triangle midsegment theorem using AA similarity.
- ☐ I can use the triangle midsegment theorem to prove relationships and to solve problems.



11.8.1 Warm-Up: “Many Small Cubes”

1. Japanese architect Sou Fujimoto exhibited a sculpture of stacked and suspended aluminum cubes as a part of the art fair in the Parisian Jardins des Tuileries’ gardens.

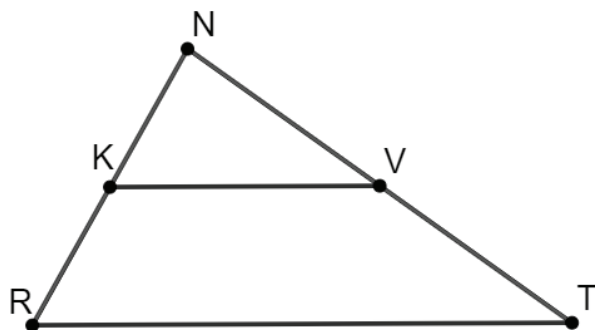


- a. What do you notice about the structure?
- b. What do you wonder?



11.8.2 Exploration: Triangle Midsegment Theorem

$\triangle NKV$ and $\triangle NRT$ are shown, where $\overline{KV} \parallel \overline{RT}$.



1. Discuss with your partner what can be concluded from the given information. Justify your conclusions.
2. Ask a classmate for two conclusions they made. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conclusion	My Summary of Their Reason	Initials



3. Given: $\triangle NRT$ with $\overline{KV} \parallel \overline{RT}$

Prove: $\triangle NKV \sim \triangle NRT$

Complete the flowchart proof.

	→	
_____		_____
		↓

4. What other information about triangles NKV and NRT can be determined from knowing that $\triangle NKV \sim \triangle NRT$?

5. Complete the table using the information $KV = 5$, $NR = 6$, $NK = 3$, and $NT = 8.8$.

$RT =$	$NV =$
$KR =$	$VT =$

6. What do you notice about the location of $\overline{KV}$?
7. What do you notice about the lengths of $\overline{KV}$ and $\overline{RT}$?



Triangle Midsegment Theorem

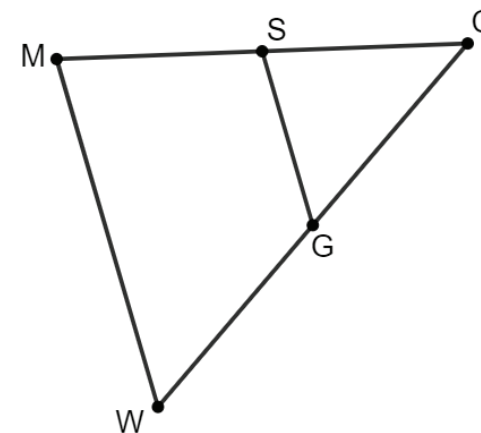
The midsegment of a triangle joins the midpoints of two sides of a triangle such that its length is half the length of the third side of the triangle, and it is parallel to the third side of the triangle.



11.8.3 Guided Practice: Triangle Midsegment Theorem

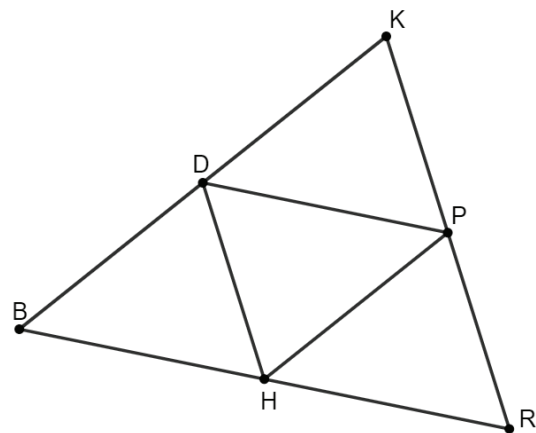
1. $\triangle MCW$ is shown, where $\overline{SG}$ is the midsegment.

$$\begin{aligned} MS &= \left(\frac{1}{2}x + 4\right), \\ GC &= (11z - 2), \\ SG &= (2y + 5), \\ MW &= (8y - 2), \\ SC &= \left(\frac{3}{4}x - \frac{1}{4}\right), \\ WG &= (9z + 1). \end{aligned}$$



- Find the value of x .
- Find the value of y .
- Find the value of z .
- What is the perimeter of $\triangle MCW$?

2. $\triangle BKR$ is shown, where D , P , and H are the midpoints and the perimeter of $\triangle DPH$ is 23.5. $DP = 17x$, $PH = (6x + 5)$, and $HD = 20x - 3$.

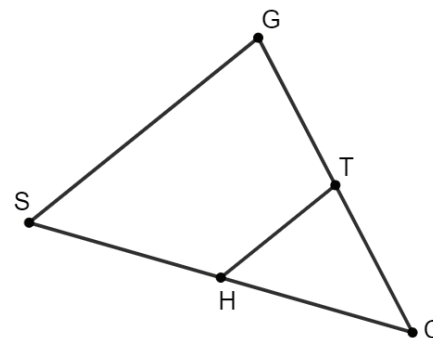


a. What is the value of x ?

b. Complete the table.

$BK =$	$DH =$
$BR =$	$KP =$
$DP =$	$KR =$

3. $\triangle SGC$ and $\triangle HTC$ are shown.



Given: $SH = HC$ and $GT = TC$

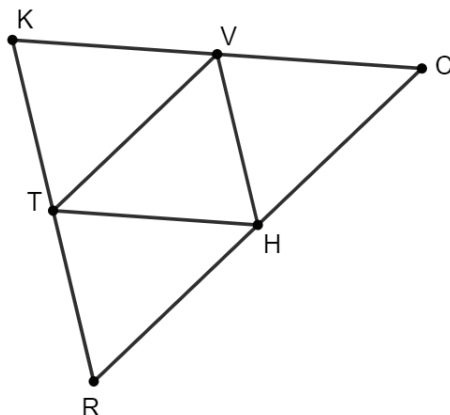
Prove: $\overline{SG} \parallel \overline{HT}$

Since $SH = HC$ and $GT = TC$, point H is the _____ of $\overline{SC}$ and point T is the _____ of $\overline{GC}$ by _____. Therefore, $\overline{HT}$ is the _____ of $\triangle SGC$ by definition and $\overline{SG} \parallel \overline{HT}$ by the _____.



11.8.4 Your Turn!

$\triangle KCR$ is shown, where T , V , and H are midpoints.
 $KC = (5x)$, $VH = (\frac{1}{3}y + 11)$, $VT = (6w - 5)$, $KR = (2y - 2)$,
 $RC = (8w + 8)$, and $TH = (3x - 4)$.



- Find the value of each variable.

$x =$	$y =$	$w =$
-------	-------	-------

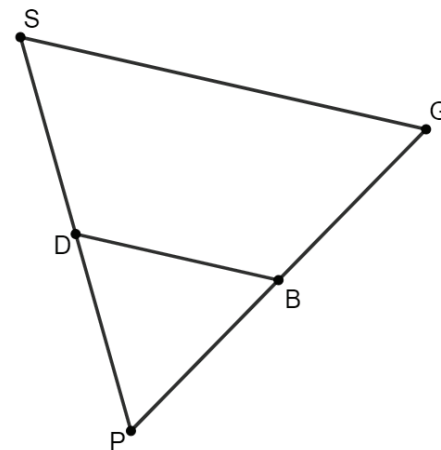
- Complete the table.

$TH =$	$KC =$
$VH =$	$KR =$
$VT =$	$RC =$



11.8.5 Wrap-Up: Error Analysis

- $\triangle SGP$ is shown, where $\overline{DB}$ is a midsegment.
 $SG = 18x + 22$ and
 $DB = 11x + 7$.



Priscilla was finding the value of x . Her work is shown.

Priscilla's Work

$$\begin{aligned} 18x + 22 &= 11x + 7 \\ 7x &= -15 \\ x &= -\frac{15}{7} \end{aligned}$$

- Write a note to Priscilla explaining her mistake.

- Find the value of x .

Unit 11, Lesson 9: Solving Problems Using Triangle Relationships



Benchmark

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.



Learning Target

- ☐ I can solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.



11.9.1 Warm-Up: Would You Rather?

1. Would you rather use a coupon worth ...

\$20 off your
entire purchase

OR

25% off your
entire purchase?

2. Explain your choice.



11.9.2 Activity: Triangle Relationships

1. For each problem, record your work and answer in the table.

Problem Letter	Work	Answer
A		
B		
C		

Problem Letter	Work	Answer
D		
E		
F		
G		



11.9.3 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

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